

Project Proposal RealEye

CMSI 4071 / Charlize, Tara, Anastasia, Anastasia, Noa

Description

RealEye is a browser-based detection tool designed to help internet users, particularly older adults and people with limited technical experience, recognize potentially AI-generated content. As generated images and text become increasingly realistic, users may unknowingly trust fake listings, impersonation attempts, or misleading messages and images. Through a Chrome extension, RealEye will scan content on the current webpage and provide a clear, accessible explanation of the available evidence.

The system will combine multiple detection methods rather than relying on a single classifier. It will check for available provenance signals, such as C2PA Content Credentials and accessible AI watermarks, and analyze text and images using pretrained and fine-tuned machine-learning models. It will then provide an AI-likelihood score and an easy-to-understand warning.

Potentially generated content will be highlighted directly on the webpage, and users can view additional information explaining why it was flagged. Its core features will include scanning webpage content, detecting potentially AI-generated text and images, generating an AI-likelihood score, checking for available watermark and provenance signals, highlighting suspicious content directly on the webpage, and providing additional information about why content was flagged. A Trusted Circle feature will also allow users to share suspicious content and its analysis with a family member or another trusted person for a second opinion.

Existing technologies such as SynthID and C2PA primarily focus on watermarking content or documenting a file's origin. Other AI detectors often return only a score and require users to copy content into a separate website. RealEye differs by bringing layered analysis directly into the browsing experience, presenting each detection signal transparently, prioritizing readable warnings and guided steps for older adults, and having the Trusted Circle feature.

The core product will use React, Chrome Extension APIs, a REST API, Python, PyTorch, and Hugging Face Transformers. Users will need a computer with Chrome and an internet connection. The initial scope covers text and image detection; video detection, phishing analysis, fact-checking, and user feedback are stretch goals.

Justification

RealEye is appropriate for CMSI 4071 because it combines concepts learned across computer science courses, including web development, APIs, software architecture, artificial intelligence, machine learning, databases, and interface design. Each team member can contribute to a meaningful part, such as the extension, frontend, backend API, provenance checks, machine-learning pipeline, evaluation process, or Trusted Circle feature. The team already has experience with React, Python, and machine-learning coursework, giving us a strong foundation for development.

The project will extend the team's knowledge in challenging areas such as browser-extension development, fine-tuning pretrained models, content-provenance standards, confidence-score interpretation, accessibility design, and integrating multiple components into one workflow. Our goal is to build a reliable end-to-end pipeline and to evaluate its performance using held-out data.

The project is achievable within one semester because its core scope is limited to a working Chrome extension that detects text and images. Additional media types and scam-detection features are stretch goals only. The team is personally invested, since AI-generated scams and misinformation increasingly affect us and our peers.